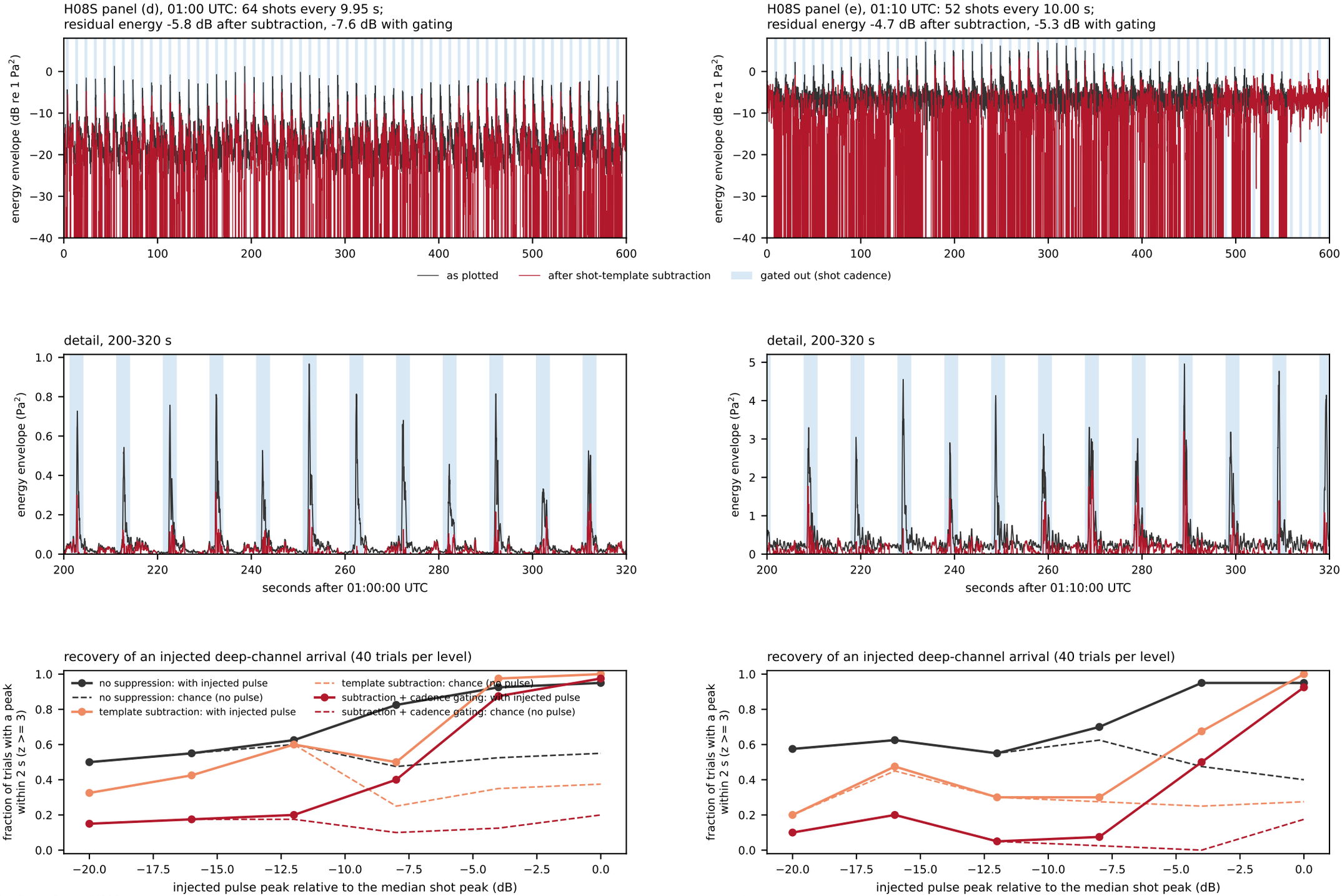
Airgun shot suppression on the digitised H08S traces lowers the residual energy by 5-8 dB and cuts chance coincidences from about 0.5 to 0.1-0.2 per trial; an added long-range arrival 4 dB below the shots is still recovered



(i) The Diego Garcia (H08S) traces are dominated by a seismic survey's airgun shots every 10 s. Top: the energy envelope as plotted (black) and after subtracting each shot's local average shape (red); blue bands mark times gated out on the shot rhythm. Middle: a two-minute detail. Bottom: how often an added long-range arrival is found (solid) against how often the same rule fires by chance at that time with nothing added (dashed).

(ii) Prereg 78c7e88 (test set c). Panels d, e: PDF vector polylines resampled to 40 Hz (as h08s_shot_outliers.load); envelope p^2 boxcar 0.25 s; shots = h08s_pulse_energy rule; local median template of 12 neighbours over [-3, +6] s scaled to the shot peak; cadence = median 9-11 s interval, gate +-1.5 s (30 % of time). z against the median/MAD of the raw envelope in shot-free gaps. Injected arrival: KRAKEN 3,703.5 km template (tau 1 s) envelope, 6 levels x 40 onsets. Digitised data; caveats C1, C2.